# A uniform interval bound for inert-prime mass

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Independently cross-checked by another model; see Section 8 for scope and subsequent clarifications.

**Abstract**

We give an elementary determinant proof of a uniform bound on the weighted mass of inert primes in arbitrary intervals. For every primitive nonprincipal real character $\chi$ of conductor $q$, we prove

$$\sum_{\substack{Y<p\le X\\ \chi(p)=-1}} \frac{\log p}{p} \le \frac{1}{2}\log\frac{X}{Y} + \frac{1}{4}\log q + \log(\log q + 33) + 18 \qquad (1 \le Y \le X).$$

For a fixed quadratic field, the allowance above $\frac{1}{2}\log(X/Y)$ depends only on the field, independently of both endpoints. Thus the same estimate controls every interval $(Y, AY]$ of a fixed ratio $A > 1$, however large $Y$ becomes.

The proof combines divided derivatives with factorial divisibility at split primes. Their contributions cancel a loss that would otherwise grow with $\log(X/Y)$, producing an explicit error independent of the interval endpoints. As a consequence, we bound the logarithmic derivative of the associated Dirichlet $L$-function: $L'(\sigma,\chi)/L(\sigma,\chi) \le \frac{1}{2}\log q + 3\log(\log q + 33) + 45$ for every real $\sigma \ge 1$. Finally, an explicit continuous model shows that the leading interval inequalities, together with specified positivity and counting constraints, are compatible with a residue of order $1/\log q$. This identifies a limitation of those aggregate constraints alone.

## 1 Introduction

Motivated by the broader aims of Tao's challenge [7], we turn from class number one to all quadratic characters. Let $\Delta$ be a fundamental discriminant, $q = |\Delta| \ge 3$, and $\chi = \chi_\Delta$ its primitive real character. Both signs of $\Delta$ are allowed. Write $t = \log q$ and

$$I_\chi(Y,X) = \sum_{\substack{Y<p\le X\\ \chi(p)=-1}} \frac{\log p}{p}, \qquad C_\chi(X) = \sum_{p\le X} \chi(p)\frac{\log p}{p}.$$

All logarithms are natural.

**Theorem 1.** *For $1 \le Y \le X$,*

$$I_\chi(Y,X) \le \frac{1}{2}\log\frac{X}{Y} + \frac{t}{4} + \log(t+33) + 18. \tag{1}$$

Fix a quadratic field and a ratio $A > 1$. Theorem 1 gives the same upper bound for the weighted sum of inert primes in $(Y, AY]$, however large $Y$ becomes. The allowance above $\frac{1}{2}\log A$ depends only on the field. Thus the estimate places a uniform restriction on the concentration of inert primes across all multiplicative scales. For narrow intervals the error may dominate; the restriction becomes informative for sufficiently wide intervals.

### 1.1 Classical uniformity and the explicit refinement

Endpoint independence for a fixed field already follows from classical analytic number theory. The prime number theorem in arithmetic progressions with its classical error term [8, Chapter 11], followed by partial summation, gives

$$I_\chi(1,x) = \tfrac{1}{2}\log x + c_\chi + o_\chi(1) \qquad (x \to \infty).$$

The remainder is bounded for $x \geq 1$. Subtracting the expressions at $X$ and $Y$ therefore gives, uniformly for $1 \leq Y \leq X$,

$$I_\chi(Y,X) = \tfrac{1}{2}\log(X/Y) + O_\chi(1).$$

The hidden constant may depend on the field. Theorem 1 supplies the explicit allowance $t/4+\log(t+33)+18$ uniformly as the field varies, by an elementary determinant argument. Its contribution is this quantitative refinement within the determinant method; qualitative endpoint independence is not claimed as new.

The original polynomial argument is due to OpenAI [1]. Green's derivative determinant [2] provides the local row operation. Van Dijk [3] proves nonvanishing by reduction at an inert prime and gives the cumulative estimate $I_\chi(1,X) \leq \frac{1}{2}\log X + \frac{1}{4}\log q + \log\log(3X) + O(1)$. Kominers [4] extends cumulative prime-mass bounds to every Frobenius class in a Galois extension. Theorem 1 is narrower in its field scope, but controls an arbitrary interval with an error independent of its location and length. It is not obtained by subtracting two upper bounds.

Our v7 proof [5], Section 4.3, retains factorial divisibility at split primes to strengthen the inert-prime comparison. Here those factors have a structural role: they remove a coefficient loss that would otherwise grow with $\log(X/Y)$. Divided derivatives and their factorial valuation cost are classical. We use them with the number of nodes proportional to $R/Y$, where $R$ is the last inert prime in the interval. The refinement studied here is this combination and its explicit interval consequence; no novelty is asserted for the individual ingredients. The source comparison does not establish priority for the exact numerical bound over all existing literature.

### 1.2 Analytic comparisons and consequences

The associated Dirichlet $L$-function is initially defined by

$$L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \qquad \operatorname{Re} s > 1.$$

Here $L'$ denotes differentiation with respect to $s$. Its logarithmic derivative $L'/L$ measures the rate of change of $L$ relative to its value; for real $\sigma > 1$ it equals $\frac{d}{d\sigma}\log L(\sigma,\chi)$.

The quasi-Riemann hypothesis (QRH) supplies stronger asymptotic control. Taking a zero-free half-plane with fixed boundary $1/2 \leq \alpha < 1$ as input—for example, the OpenAI result with boundary $\alpha = 11/12$ [9, Theorem 1.1]—the standard explicit formula gives

$$\vartheta_\chi(x) := \sum_{p \leq x} \chi(p)\log p \ll_\alpha x^\alpha \log^2(qx), \qquad x \geq 2,$$

uniformly in $q$ and $\chi$; see [8, Section 11.1] for the explicit formula and the uniform zero count. Here all nontrivial zeros have real part at most $\alpha < 1$; the functional equation bounds their real parts below by $1-\alpha$, and removing prime powers changes the estimate by $O(\sqrt{x}\log^2 x)$. For clarity, the resulting interval comparison is

$$I_\chi(Y,X) \leq \tfrac{1}{2}\log(X/Y) + O_\alpha(\log\log(3q)) \qquad (1 \leq Y \leq X).$$

Indeed, put $U = (\log(3q))^{3/(1-\alpha)}$. Mertens' estimate bounds the absolute value of $C_\chi(\min(X,U))$ by $O_\alpha(\log\log(3q))$. For $X \geq U$, partial summation of the displayed estimate for $\vartheta_\chi$ gives

$$|C_\chi(X) - C_\chi(U)| \ll_\alpha U^{\alpha-1}\log^2(qU) = O_\alpha(1).$$

Thus $C_\chi(X) = O_\alpha(\log\log(3q))$ uniformly in $X$. Combining the difference $C_\chi(X) - C_\chi(Y)$ with Mertens' estimate, and discarding the nonnegative ramified-prime term, proves the comparison. This is a deduction from the cited analytic input. Its dependence on $q$ is asymptotically stronger than Theorem 1; we do not compute a numerical constant for it here. The value of the present argument is its explicit estimate and elementary mechanism, rather than a stronger asymptotic prime-distribution theorem.

Cairn [6], Appendix C, already treats all quadratic characters and transfers the determinant profile to the residue and regular part of $\zeta(s)L(s,\chi)$. In particular, that paper proves

$$\frac{L'(1,\chi)}{L(1,\chi)} < \frac{t}{2} + 8\log t + 45 \qquad (t \geq 10).$$

Our interval bound gives the following replacement, without importing the ideal count or a preliminary lower bound for $L(1,\chi)$.

**Corollary 2.** *Put $B_q = t/2 + 3\log(t+33) + 45$. For $1 \leq Y \leq X$,*

$$C_\chi(X) - C_\chi(Y) \geq -B_q. \tag{2}$$

*For every real $\sigma \geq 1$, $L(\sigma,\chi) > 0$ and*

$$\frac{L'(\sigma,\chi)}{L(\sigma,\chi)} \leq B_q. \tag{3}$$

These statements do not assert a new class-number classification, a fixed zero-free half-plane, or an improvement on the strongest available bounds obtained using that heavier input. The proof below uses elementary prime estimates, quadratic arithmetic, and finite determinants. The $L$-function corollary additionally uses the Euler product and convergence of the Dirichlet series and its derivative on the real line at 1. No zero-free theorem is an input.

## 2 Elementary prime estimates

Put $c = 2\log 2 < 7/5$, $M(x) = \sum_{p\leq x}(\log p)/p$, and $\vartheta(x) = \sum_{p\leq x}\log p$. We will use, for real $x \geq 1$,

$$\vartheta(x) \leq cx, \qquad \log x - 4 \leq M(x) \leq \log x + c. \tag{4}$$

Consequently, for $1 \leq a \leq b$ and $a > 1$ in the last inequality,

$$|M(b) - M(a) - \log(b/a)| < 6, \tag{5}$$

$$\sum_{p>a}\frac{\log p}{p^2} \leq \frac{2c}{a}, \qquad \sum_{p>a}\frac{\log p}{p(p-1)} \leq \frac{2c}{a-1}. \tag{6}$$

For completeness, the usual binomial-coefficient induction gives $\vartheta(n) \leq 2n\log 2$: for $n = 2m$ use $\binom{2m}{m}$ and the primes in $(m, 2m]$; for $n = 2m+1$ use $\binom{2m+1}{m} \leq 2^{2m}$ and $(m+1, 2m+1]$. The factorial identity gives

$$nM(n) - \vartheta(n) \leq \log(n!) \leq n\sum_{p\leq n}\frac{\log p}{p-1}, \qquad \log(n!) \geq n\log n - n.$$

Partial summation of $\vartheta(x) \le cx$ gives (6). Retaining the boundary term at 2 also gives

$$\sum_p \frac{\log p}{p(p-1)} \le \frac{\log 2}{2} + \frac{3}{2}\left(c - \frac{\log 2}{4}\right) = \frac{25}{8}\log 2.$$

Thus $M(n) \ge \log n - 1 - (25/8)\log 2$. Taking $n = \lfloor x \rfloor$ and using $1 + (33/8)\log 2 < 4$ proves the lower bound in (4); the upper bound follows from $\log(n!) \le n \log n$. The range $1 \le x < 2$ is immediate. For example, $\log 2 < 7/10$ follows by retaining the first four terms of the exponential series for $e^{7/10}$.

## 3 A determinant adapted to an interval

Fix for now an integer $K \ge 32$. If the interval contains an inert prime, let $R$ be its largest one. Put

$$Z = R/Y > 1, \qquad n = \lceil Z \rceil, \qquad S = \lfloor n/K \rfloor, \qquad J = \lceil RS/n \rceil. \tag{7}$$

We first assume

$$n \ge 2K, \qquad n \ge \sqrt{q}. \tag{8}$$

Then $n \le R$, $S \ge 2$, $R \ge 64$, and $J < R$.

Choose exactly $n$ points $z = u + v\sqrt{\Delta}$ from the grid

$$0 \le u < \lceil \sqrt{n} q^{1/4} \rceil, \qquad 0 \le v < \lceil \sqrt{n} q^{-1/4} \rceil,$$

and put $w = \bar{z}$. There are at least $n$ grid points, and both embeddings have modulus less than

$$H = 2\sqrt{n} q^{1/4}, \qquad \log H = \log 2 + \tfrac{1}{2}\log n + \tfrac{1}{4}t. \tag{9}$$

The coordinate ranges have lengths at most $R$, since $q^{1/4}\sqrt{n} \le n \le R$. As $R$ is odd and inert, $\mathcal{O}_{\mathbb{Q}(\sqrt{\Delta})}/R\mathcal{O} \simeq \mathbb{F}_{R^2}$ and $1, \sqrt{\Delta}$ are an $\mathbb{F}_R$-basis. The selected points remain distinct after reduction.

Use rows $z^r w^s$, $0 \le r < R$, $0 \le s < S$, and at each point use the $J$ divided derivatives

$$D_z^{[k]}(z^r w^s) = \binom{r}{k} z^{r-k} w^s, \qquad 0 \le k < J,$$

with value zero for $k > r$. The entries are algebraic integers.

**Lemma 3.** *This matrix has full row rank $N = RS$.*

*Proof.* Modulo $R$, Frobenius gives $w = z^R$. A nonzero row combination therefore becomes a nonzero polynomial with distinct exponents $r + Rs$ and degree at most $RS - 1$. Its derivatives of orders $k < J < R$ agree with the prescribed derivatives after substitution; equivalently, multiply the divided derivatives by the invertible $k!$. Vanishing of all the columns would give $n$ distinct roots, each of multiplicity at least $J$, whereas $nJ \ge RS$. This is van Dijk's argument [3], with the different node and derivative counts in (7). □

Select a nonzero $N$-square minor $d$ and put $\mathcal{D} = |\,\mathrm{N}_{\mathbb{Q}(\sqrt{\Delta})/\mathbb{Q}}\, d| \in \mathbb{Z}_{>0}$. Define the total row degrees and the selected column orders by

$$U = \frac{N(R-1)}{2}, \qquad W = \frac{N(S-1)}{2}, \qquad V = \sum_{\text{selected columns}} k,$$

and set

$$b = \frac{W}{U}, \qquad v = \frac{V}{U}, \qquad d_0 = 1 - b - v. \tag{10}$$

The symbol $d_0$ is a real coefficient, not a discriminant or determinant. The parameter choices imply

$$0 \le b \le \frac{1}{K}, \qquad 0 \le v < \frac{3}{K}, \qquad d_0 > 1 - \frac{4}{K} \ge \frac{7}{8}. \tag{11}$$

Indeed, $(S-1)/(R-1) \le S/R \le 1/K$, and $J - 1 < RS/n \le R/K$, so $v < 2R/[K(R-1)] < 3/K$.

### 3.1 Size at both embeddings

For every nonzero determinant term, the sum of the exponents $r + s - k$ is $U + W - V$. The bound $\binom{r}{k} \le 2^r$ therefore gives

$$\log \mathcal{D} \le 2\log(N!) + 2U\log 2 + 2(U + W - V)\log H.$$

Since $N \le R^2$ and $R \ge 64$, $\log(N!)/U \le 4\log R/(R-1) < 1$. Thus

$$\frac{\log \mathcal{D}}{2U} \le (1 + b - v)\log H + \log 2 + 1. \tag{12}$$

The absence of a $J \log R$ term is the reason for dividing the derivative columns by their factorials. The corresponding arithmetic cost must still be paid, as follows.

## 4 Inert and split primes in the same minor

Put $Q = R/S$. Then $K \le Q < R$ and $Y < Q$. Write $I = I_\chi(Q, R)$ and let $P$ be the analogous sum over split primes in $(Q, R]$.

### 4.1 Inert primes and the factorial cost

At an inert prime $p$, Green's integral unitriangular row operation is

$$z^r w^s \longmapsto z^{r-pa}(z^p - w)^a w^s, \qquad a = \min(\lfloor r/p \rfloor, S - 1 - s).$$

Every ordinary derivative of $z^p - w$ is divisible by $p$ after evaluation, including its zeroth derivative by Frobenius. The ordinary derivative minor has norm divisible by $p^{2E_p}$, where

$$E_p = \sum_{r<R, s<S} \min(\lfloor r/p \rfloor, S - 1 - s).$$

Dividing its selected columns by $k!$ gives exactly $d$, hence

$$\frac{v_p(\mathcal{D})}{2U} \ge \frac{E_p}{U} - \frac{v}{p-1}. \tag{13}$$

We used $\sum_k v_p(k!) \le V/(p-1)$, and do not claim that the ordinary-derivative divisibility survives unchanged.

For every integer $a \ge 0$, $\sum_{s<S} \min(a, S-1-s) \ge Sa - a(a+1)/2$. Using $r/p - 1 \le \lfloor r/p \rfloor \le r/p$ gives

$$\frac{E_p}{U} \ge \frac{1}{p} - \frac{2}{R-1} - \frac{1}{2Sp} - \frac{2R-1}{6Sp^2}. \tag{14}$$

### 4.2 Split primes

At a split prime replace $w^s$ by $(w)_s = w(w-1)\cdots(w-s+1)$. This is another integral unitriangular change preserving the same selected minor. Each block of $p$ consecutive factors vanishes in both components of $\mathcal{O}/p\mathcal{O} \simeq \mathbb{F}_p \times \mathbb{F}_p$. Thus its values belong to $p^{\lfloor s/p \rfloor}\mathcal{O}$. Divided differentiation in $z$ preserves this factor and integrality. Consequently

$$\frac{v_p(\mathcal{D})}{2U} \geq \frac{\sum_{r,s} \lfloor s/p \rfloor}{U} \geq \frac{b}{p} - \frac{2}{R-1}. \tag{15}$$

The inert and split operations are used at disjoint sets of primes, so their contributions can be added. This is the factorial mechanism from [5, Section 4.3].

### 4.3 A uniform bound for all discarded terms

Sum (13)–(15) over $(Q, R]$. Equations (4)–(6) show that the total error in retaining $(1-v)I + bP$ is at most

$$\frac{2cR}{R-1} + \frac{2c}{3} + \frac{\log S + 6}{2S} + \frac{6c}{K(Q-1)} < 6. \tag{16}$$

The four terms respectively bound the floor loss across the disjoint prime sets, the $p^{-2}$ row loss, the $I/(2S)$ term, and the difference $1/(p-1) - 1/p$ in the column cost. For the third term, $I \leq M(R) - M(Q) < \log S + 6$. For $R \geq 64$, $S \geq 2$, and $Q, K \geq 32$, the four bounds are, respectively, less than 2.845, 0.934, 1.675, and 0.009. The sum is less than 6. All inequalities are uniform in $Y$.

Combining this with (12) and $1 + \log 2 + 6 < 8$ gives

$$(1-v)I + bP \leq (1 + b - v)\log H + 8. \tag{17}$$

Let $A$ be the ramified-prime mass in $(Q, R]$. Then

$$I + P = M(R) - M(Q) - A, \qquad A \leq t/Q.$$

Substitution in (17), and $M(R) - M(Q) \geq \log S - 6$, prove

$$d_0 I \leq (1 + b - v)\log H - b\log S + 6b + bt/Q + 8. \tag{18}$$

Crucially, $1 + b - v = d_0 + 2b$. The $b\log n$ contributed by the height in (9) cancels against $b\log S$, up to $b\log(n/S)$. Discarding the split-prime term would lose this cancellation and leave an error proportional to $\log(X/Y)/K$.

## 5 Completing the constants

Dividing (18) by $d_0$ gives

$$\begin{aligned} I \leq {} & \tfrac{1}{2}\log n + \tfrac{1}{4}t + \log 2 + \frac{8}{d_0} \\ & + \frac{b}{d_0}\left(\frac{t}{2} + 2\log 2 + \log(n/S) + 6 + \frac{t}{Q}\right). \end{aligned} \tag{19}$$

The omitted interval has mass at most

$$I_\chi(Y, Q) \leq \log(Q/Y) + 6 \leq \log(n/S) + 6.$$

Here $R \le nY$, $Y < Q$, and $Q = R/S$. Also $S \ge n/(2K)$, $n \le 2Z$, and

$$\frac{b}{d_0} \le \frac{1}{K-4}, \qquad \frac{8}{d_0} \le \frac{8K}{K-4}, \qquad Q \ge K.$$

It follows that

$$I_\chi(Y,R) \le \frac{1}{2}\log Z + \frac{t}{4} + \log K + \frac{K+2}{2K(K-4)}t + C_K, \tag{20}$$

where

$$C_K = \frac{5}{2}\log 2 + 6 + \frac{8K}{K-4} + \frac{2\log 2 + \log(2K) + 6}{K-4} < \frac{35}{2}. \tag{21}$$

For $K \ge 32$ the two variable terms decrease; the derivative of the last one has numerator $(K-4)/K - [3\log 2 + \log K + 6] < 0$. At 32 the upper bound $\log 2 < 7/10$ already gives $C_K < 17.308 < 35/2$.

Choose

$$K = \lceil t \rceil + 32.$$

Then $K < t + 33$ and

$$\frac{K+2}{2K(K-4)}t \le \frac{(K-32)(K+2)}{2K(K-4)} < \frac{1}{2}.$$

Equations (20)–(21) give (1) with $R$ in place of $X$ in the nontrivial range.

If (8) fails, use $I_\chi(Y,R) \le \log Z + 6$. If $n < \sqrt{q}$, then $\frac{1}{2}\log Z \le t/4$. If $n < 2K$, then $\frac{1}{2}\log Z < \frac{1}{2}\log(2K) \le \log K$. Either case gives the same claimed bound, with room to spare. If there is no inert prime in the interval, the assertion is immediate. Finally, $I_\chi(Y,X) = I_\chi(Y,R)$ and $R \le X$, so the right side can only increase when $R$ is replaced by $X$. This proves Theorem 1.

## 6 The signed profile and the logarithmic derivative

The total ramified mass satisfies

$$\sum_{p \mid q} \frac{\log p}{p} \le M(t+1) + \frac{t}{t+1} < \log(t+33) + \frac{12}{5}. \tag{22}$$

Subtract twice (1) and this ramified contribution from $M(X) - M(Y) \ge \log(X/Y) - 6$. The result is

$$C_\chi(X) - C_\chi(Y) \ge -\frac{t}{2} - 3\log(t+33) - \frac{222}{5} > -B_q,$$

which proves (2).

For $\eta > 0$, summation by parts gives

$$\sum_p \frac{\chi(p)\log p}{p^{1+\eta}} = \eta \int_1^\infty C_\chi(x) x^{-1-\eta}\, dx \ge -B_q.$$

The higher-prime-power terms in the Euler logarithmic derivative are nonnegative after summing at each prime:

$$\sum_{a \ge 2} \frac{\chi(p)^a}{p^{a\sigma}} = \frac{\chi(p)^2 p^{-2\sigma}}{1 - \chi(p)p^{-\sigma}} \ge 0.$$

It follows that $L'(\sigma,\chi)/L(\sigma,\chi) \le B_q$ for $\sigma > 1$. Positivity at 1 need not be imported: integrate from $1+\varepsilon$ to 2 to obtain $L(1+\varepsilon,\chi) \ge e^{-B_q} L(2,\chi) > 0$. The bounded partial sums of the nonprincipal

periodic character ensure continuity of $L$ and $L'$ on the real line at 1. Taking the limit gives $L(1,\chi) > 0$ and (3).

The stronger interval statement also gives, for every $Y \geq 1$,

$$\sum_{p>Y} \frac{\chi(p)\log p}{p^{1+\eta}} \geq -B_q Y^{-\eta}.$$

Indeed, integrate $C_\chi(x) - C_\chi(Y)$ from $Y$ to infinity. This conclusion concerns the signed tail, not its absolute value.

**Corollary 4.** *For every $Y \geq 1$ there is a split prime in*

$$Y < p \leq e^{53}\sqrt{q}\,(\log q + 33)^4 Y.$$

*Proof.* The split-prime mass in $(Y, X]$ is at least

$$\frac{1}{2}\log(X/Y) - \frac{t}{4} - 2\log(t+33) - \frac{132}{5},$$

by subtracting (1) and (22) from (5). At the stated endpoint this is $1/10 > 0$. □

This illustrates the uniformity in the starting point. It is an application of this proof, not a claim to improve on the strongest available analytic prime-distribution theorems.

For comparison with the general residue bound in [6, Appendix C.4, (C13)], importing that paper's Fourier ideal count and regular-part transfer now gives, for $t \geq 10$,

$$L(1,\chi) > \frac{\zeta(2) - 21/t^3}{t + 8\log t + 3\log(t+33) + 46}. \tag{23}$$

This is a corollary relative to those stated inputs, not an additional self-contained proof of the ideal count. To check the substitution, take $X_0 = \sqrt{q}\,t^8$ in (C13), use (3), and use $\gamma - 1 < 0$. The previous denominator was $t + 16\log t + 46$. We do not claim a better limiting coefficient than Cairn's stronger moment and Burgess refinements.

## 7 A limit of the aggregate information

It is tempting to expect that combining enough scale inequalities will automatically replace $1/\log q$ by $1/\log\log q$ in the residue bound. The following explicit model shows why a specified collection of relaxed constraints cannot establish that improvement. It is not asserted to arise from actual Dirichlet characters.

Use the scaled variable $u = \log x/t$. Let the total prime-log density be 1, the inert density be

$$i(u) = \frac{1 + e^{-2u}}{2},$$

and the split density be $(1 - e^{-2u})/2$. These densities are nonnegative and sum to 1. On every interval $0 \leq a \leq b$,

$$\int_a^b i(u)\,du = \frac{b-a}{2} + \frac{e^{-2a} - e^{-2b}}{4} \leq \frac{b-a}{2} + \frac{1}{4}. \tag{24}$$

Thus the leading part of Theorem 1 holds on *every* interval, not only at a single cutoff.

To retain multiplicative information as well, define the positive primitive-prime logarithmic measure

$$d\nu(u) = \frac{1 - e^{-2u}}{u}\, du \qquad (u > 0).$$

For $z > 0$, the elementary identity

$$\int_0^\infty \frac{e^{-zu} - e^{-(z+2)u}}{u}\, du = \int_z^{z+2} \frac{dv}{v} = \log \frac{z+2}{z}$$

shows that its convolution exponential has Laplace transform

$$F(z) = \exp\left(\int_0^\infty e^{-zu}\, d\nu(u)\right) = 1 + \frac{2}{z}. \tag{25}$$

Equivalently, the resulting primitive-ideal measure is $\mu = \delta_0 + 2\, du$. It has the identity ideal, positivity, and an exactly constant tail density at every positive scale.

We specify the counting constraints retained in this relaxation. The arithmetic quantity being modeled is the weighted measure

$$\mu_\chi = \sum_{n \geq 1} \frac{b_\chi(n)}{n}\, \delta_{\log n/t}, \qquad \frac{\zeta(s)L(s,\chi)}{\zeta(2s)} = \sum_{n \geq 1} \frac{b_\chi(n)}{n^s},$$

where $b_\chi(n) \geq 0$ counts primitive ideals and $b_\chi(1) = 1$. For a threshold $a \in \{1/2, 1/4\}$ and a normalized residue $r > 0$ corresponding to $tL(1,\chi)/\zeta(2)$, the retained conditions on a locally finite measure $\mu$ on $[0,\infty)$ are

$$\mu - \delta_0 \geq 0, \tag{26}$$

$$\mu((u,v]) = r(v-u) \quad (a < u < v), \tag{27}$$

$$1 \leq \mu([0,a]) \leq r(a + \tfrac{1}{2}). \tag{28}$$

Here (27) retains the leading constant density beyond the counting threshold: $a = 1/2$ for the finite-Fourier count, and $a = 1/4$ when the Burgess count is imported [6, Appendices C.5–C.6, (C15) and (C22)]. The lower bound in (28) is the identity ideal. The upper bound retains the regular-part constraint from $L'(1,\chi)/L(1,\chi) \leq t/2 + o(t)$. Indeed, as $z \downarrow 0$,

$$\int_{[0,\infty)} e^{-zu}\, d\mu(u) = \frac{r}{z} + \mu([0,a]) - ra + O(z),$$

so the retained bound on the constant term is $\mu([0,a]) - ra \leq r/2$. These are limiting constraints; no finite-$q$ counting error term or realization by a sequence of characters is imposed.

For example, these conditions imply the first-moment bound

$$\int_{[0,V]} u\, d\mu(u) \leq a(\mu([0,a]) - 1) + \frac{r}{2}(V^2 - a^2), \qquad V \geq a,$$

by removing the identity atom and using $u \leq a$ on the head. Our model satisfies (26)–(28) with $r = 2$ for both thresholds, since

$$\mu([0,a]) = 1 + 2a = 2(a + \tfrac{1}{2}), \qquad \mu((u,v]) = 2(v-u) \quad (0 \leq u < v).$$

Its first moment is $V^2$, while the displayed upper bound is $V^2 + a^2$. Thus the counting conditions and their first-moment consequence hold explicitly.

In the usual residue scaling, $\zeta(1+z/t) \sim t/z$ and $\zeta(2+2z/t) \to \zeta(2)$. Consequently the model (25) corresponds to

$$\frac{t}{\zeta(2)} L(1+z/t) \longrightarrow z+2, \qquad tL(1) \longrightarrow 2\zeta(2), \qquad \frac{1}{t}\frac{L'(1)}{L(1)} \longrightarrow \frac{1}{2}.$$

The scaled last expression saturates the leading logarithmic derivative coefficient. Hence positivity of this convolution exponential, the constant primitive-ideal tail, and all the leading interval inequalities are compatible with a residue of order $1/\log q$.

This is a countermodel only to that limiting relaxation. It omits the existence of a single periodic character with the prescribed prime behavior, its functional equation, individual ideal-class geometry, and other arithmetic correlations. It proves neither optimality of the determinant method in general nor a limitation of every possible use of these ideas. It identifies the kind of additional information a larger improvement would have to retain.

## 8 Checks and remaining work

The accompanying Python program checks exact quadratic-ring minors, finite-field ranks, inert-prime divisibility with the column-factorial cost, split-prime row divisibility, the finite row-sum inequality, and the rational constant bounds. Its small examples deliberately use more varied derivative-to-node ratios than the final conservative parameter choice. They check the algebraic mechanisms; they do not replace the proof over the infinite range. The saved output records the executed checks and their scope. There are 25 nonzero-minor checks, 85 inert-prime and 51 split-prime valuation checks, 55,091 finite row-sum bounds, and 2,286 separately expanded row-sum identities. In 13 cases, omitting the column-factorial cost would give a false divisibility claim; the corrected bound passes.

An independent model cross-check, reported on 26 October 2026, reviewed the ten-page draft and its source and found no substantive gap in Theorem 1, Corollary 2, or Corollary 4. The checker reported reproducing all the supplied checks and verifying another 11 determinant norms through a separate implementation using ordinary integer block matrices, covering both real and imaginary fields. It also confirmed the substitution giving (23), subject to its stated external ideal-count input. The report is recorded in the accompanying `cross_check.txt`. The two presentation corrections it requested—the v7 section references and the explicit specification of the model's counting constraints—are incorporated in this revision. This is a model cross-check, not formal verification or human peer review, and it does not establish priority.

The source comparison covers classical analytic estimates, the QRH comparison, Green, van Dijk, Kominers, the relevant Cairn manuscript, and the v7 proof; it does not establish priority over all current manuscripts. The generalization to other Frobenius classes is not claimed here.

## References


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